

Test 1 Prep

If $(d > 1)$ is a linear combinations (with integer coefficients) of m and n , can you say that $d = \gcd(m, n)$??

Quiz 1 TRUE or FALSE?

(1) 1. Suppose that m and n are integers not both zero and $d \in \mathbb{Z}$, $d > 1$ such that $d = xm + yn$, where $x, y \in \mathbb{Z}$. Then $d = \gcd(m, n)$.

? $\swarrow$
True

$\searrow$?
False

Bézout's Lemma Let $m, n \in \mathbb{Z}$ not both zero. Then $d = \gcd(m, n)$ exists and $d = xm + yn$, where $x, y \in \mathbb{Z}$.

$$d = \gcd(m, n) \xRightarrow{\text{Bézout's Lemma}} d = xm + yn$$

??

$\leftarrow$

True?

$\longrightarrow$ Converse of Bézout's Lemma is true if $(d = 1)$

$$\left. \begin{array}{l} m = 4 \\ n = 8 \end{array} \right\} \gcd(m, n) = \gcd(4, 8) = 4$$

$$\underline{d = 12} = \underbrace{1}_x \cdot 4 + \underbrace{1}_y \cdot 8$$

Practice problems Ch 1, 1 Q5

$$(-1)^1 \cdot 1^2 + \dots + (-1)^n n^2 \stackrel{??}{=} \frac{1}{2} (-1)^n n(n+1), \text{ for all } n \geq 1$$

For $n \geq 1$, we define the following statement:

$$P_n: (-1)^1 1^2 + \dots + (-1)^n n^2 = \underline{\underline{\frac{1}{2} (-1)^n n(n+1)}}$$

We first check that P_1 holds:

$$(-1)^1 1^2 = -1$$

$$\frac{1}{2} (-1)^1 1(1+1) = -1$$

Thus, P_1 holds. Assume that P_k holds (for some $k \geq 1$) and let us prove that

P_{k+1} holds.

We want to show:

$$\underbrace{(-1)^1 1^2 + \dots + (-1)^k k^2}_{P_k \text{ true}} + (-1)^{k+1} (k+1)^2 \stackrel{?}{=} \frac{1}{2} (-1)^{k+1} (k+1)(k+2)$$



Using that P_k is true we get

$$(-1)^1 \cdot 1^2 + \dots + (-1)^k k^2 + (-1)^{k+1} (k+1)^2 =$$

P_k is true $\downarrow$

$\frac{1}{2} (-1)^k k(k+1)$

$$= \frac{1}{2} (-1)^k k(k+1) + (-1)^{k+1} (k+1)^2$$

$\xrightarrow{(-1)^k \cdot (-1)}$

$$= (-1)^k (k+1) \left(\frac{1}{2} k + (-1)(k+1) \right)$$

$-1 - k$

$$= (-1)^k (k+1) \frac{k - 2k - 2}{2}$$

$$\left\{ \begin{aligned} \frac{k}{2} - 1 - k &= \\ &= \frac{k - 2 - 2k}{2} \end{aligned} \right.$$

$$= (-1)^k (k+1) \frac{-k-2}{2} =$$

$$= \frac{1}{2} (-1)^k (k+1) (-1) (k+2) = \frac{1}{2} (-1)^{k+1} (k+1)(k+2)$$

Practice problems 11 (b)

Ch 1

$$\left. \begin{array}{l} 3x \equiv 2 \pmod{5} \\ x \equiv 2 \pmod{3} \end{array} \right\}$$

Since $\gcd(5, 3) = 1$, we know that (by the Chinese Remainder Thm) that a common solution exists.

Any solution of the equation $x \equiv 2 \pmod{3}$ has the form $x = 2 + 3K$, for some $K \in \mathbb{Z}$. The goal is to find K such that $2 + 3K$ is a solution of the equation $3x \equiv 2 \pmod{5}$.

$$3(2 + 3K) \equiv 2 \pmod{5} \Rightarrow 6 + 9K \equiv 2 \pmod{5}$$

$$\Rightarrow 9K \equiv -4 \pmod{5} \Rightarrow \boxed{4K \equiv 1 \pmod{5}}$$

$$\overline{9} = \overline{4} \text{ in } \mathbb{Z}_5$$

$$9 \equiv 4 \pmod{5}$$

$$-4 \equiv 1 \pmod{5}$$

$$4K \equiv 1 \pmod{5}$$

$$\overline{4}^{-1}$$

Notice that $\overline{4}$ has a multiplicative inverse in $\mathbb{Z}_5$ because $\gcd(4, 5) = 1$. We know that $\overline{1}$ and $\overline{p-1}$ are self-inverses in $\mathbb{Z}_p$ for p , prime. So in this case, $p = 5$, and so $\overline{4}^{-1} = \overline{4}$

Multiplying by 4 in the equation $4K \equiv 1 \pmod{5}$, we get that

$$K \equiv 4 \pmod{5}$$

Taking (for example) $K = 4$, we get that $\underline{x = 2 + 3 \cdot K = 2 + 3 \cdot 4 = 14}$ is a common solution. ↑

Any other ^{common} solution is congruent
to 14 modulo $5 \cdot 3 = 15$, so
all the solutions are of the form

$$14 + 15m, \quad m \in \mathbb{Z}$$

Question 10 Practice problems ch 1

$$n = d_j d_{j-1} \dots d_2 d_1 d_0$$

$$n = 5751 = \underbrace{1}_{d_0} + \underbrace{5}_{d_1} \cdot 10 + \underbrace{7}_{d_2} \cdot 10^2 + \underbrace{5}_{d_3} \cdot 10^3$$

$$3 \mid n \iff 3 \mid (d_0 + d_1 + \dots + d_j) \quad \leftarrow$$

Notice that

$$n = d_0 + d_1 \cdot 10 + d_2 \cdot 10^2 + \dots + d_j \cdot 10^j$$

$$3 \mid n \iff \bar{n} = \bar{0} \text{ in } \mathbb{Z}_3$$

$$3 \mid (d_0 + \dots + d_j) \iff \overline{d_0 + \dots + d_j} = \bar{0} \text{ in } \mathbb{Z}_3$$

$$\overline{d_0 + \dots + d_j} = \overline{d_0} + \overline{d_1} + \dots + \overline{d_j}$$

In $\mathbb{Z}_3$,

$$\overline{10} = \overline{1}$$

$$\mathbb{Z}_3 = \{\overline{0}, \overline{1}, \overline{2}\}$$

$$\overline{10^k} = \overline{1} \text{ for all } k \geq 1$$

$$\overline{a} + \overline{b} = \overline{a+b}$$

$$\overline{a} \cdot \overline{b} = \overline{ab}$$

$$a \equiv b \pmod{n}$$

$$\overline{a} \text{ in } \underline{\mathbb{Z}_n}$$

In $\mathbb{Z}_3$ we have

$$\overline{n} = \overline{d_0 + d_1 \cdot 10 + d_2 \cdot 10^2 + \dots + d_j \cdot 10^j} =$$

$$= \bar{d}_0 + \bar{d}_1 \cdot \overline{10} + \bar{d}_2 \cdot \overline{10}^2 + \dots + \bar{d}_j \cdot \overline{10}^j =$$

$$= \bar{d}_0 + \bar{d}_1 + \bar{d}_2 + \dots + \bar{d}_j = \overline{d_0 + d_1 + \dots + d_j}$$

$$\underline{\bar{n} = \bar{0} \text{ in } \mathbb{Z}_3} \Leftrightarrow \overline{d_0 + d_1 + \dots + d_j} = \bar{0} \text{ in } \mathbb{Z}_3$$

$$3 \mid n \Leftrightarrow 3 \mid (d_0 + d_1 + \dots + d_j)$$

T or F

$$10^{515} \equiv 6 \pmod{7}$$
